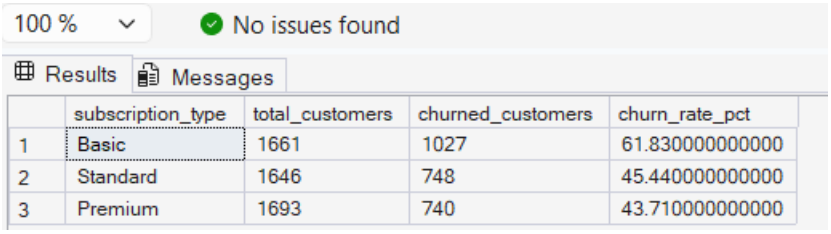
OTT Subscriber Churn & Retention Analysis

SQL Analysis Results and Business Findings

Q1. Bussines Question: Where are we losing the most customers?

Purpose: Identify regions with the highest number of churned customers.

Business Goal: Identify which subscription plans have the highest customer churn and churn rates.



	subscription_type	total_customers	churned_customers	churn_rate_pct
1	Basic	1661	1027	61.830000000000
2	Standard	1646	748	45.440000000000
3	Premium	1693	740	43.710000000000

Key Finding: Basic has the highest churn rate (61.83%) and the most churned customers (1,027). The business should investigate why Basic customers leave more frequently.

Q2: Are customers disengaging before they churn?

-- Compare login inactivity with churn status.

Business goal: Find out whether customers who haven't logged in for longer periods are more likely to be churned. This could help a retention team identify customers who may need attention.

100 % No issues found

Results Messages

	last_login_days	total_customers	churned_customers
1	0	81	3
2	1	85	10
3	2	90	10
4	3	81	12
5	4	93	12
6	5	72	14
7	6	77	10
8	7	71	16
9	8	78	21
10	9	85	19
11	10	94	27
12	11	89	20
13	12	76	24
14	13	77	21
15	14	60	19
16	15	62	17
17	16	100	21
18	17	65	18
19	18	91	21
20	19	68	17
21	20	87	29
22	21	76	24
23	22	80	32
24	23	89	25
25	24	88	37
26	25	90	22
27	26	92	31
28	27	89	35
29	28	85	30
30	29	74	31
31	30	86	33

100 % No issues found

Results Messages

	last_login_days	total_customers	churned_customers
31	30	86	33
32	31	69	47
33	32	75	55
34	33	86	66
35	34	88	65
36	35	69	45
37	36	86	59
38	37	102	75
39	38	68	51
40	39	88	67
41	40	91	66
42	41	88	66
43	42	76	56
44	43	76	57
45	44	96	83
46	45	81	67
47	46	98	76
48	47	83	63
49	48	83	64
50	49	81	67
51	50	81	59
52	51	74	57
53	52	80	60
54	53	93	65
55	54	92	73
56	55	78	56
57	56	72	53
58	57	83	63
59	58	83	62
60	59	79	60
61	60	70	51

Key finding: In the observed dataset,

- 3 of 81 customers with 0 days since their last login were churned (3.7%),
- 33 of 86 customers at 30 days (38.4%).
- This suggests an association between longer login inactivity and higher churn.

Q3 — Which customer losses have the greatest financial exposure?

Business goal : Understand which subscription plans have the highest monthly-fee value associated with churned customers, so retention efforts can consider financial exposure as well as the number of customers lost.

100 % No issues found

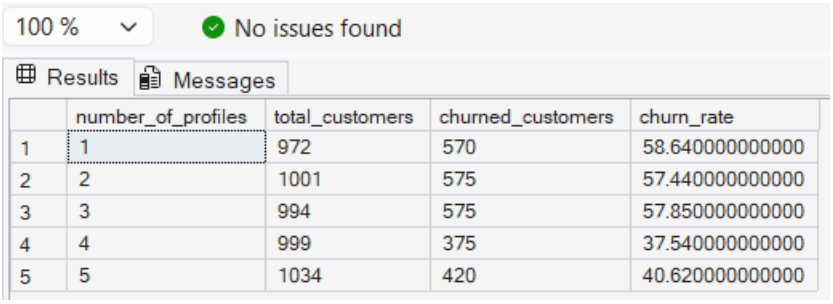
Results Messages

	subscription_type	churned_customers	monthly_fee_exposure	average_monthly_fee
1	Premium	740	13312.5998306274	17.9899997711182
2	Standard	748	10464.5198287964	13.9899997711182
3	Basic	1027	9232.72976493835	8.98999977111816

Key finding : Premium has the highest monthly-fee exposure at 13,312.60, despite having 740 churned customers compared with Basic's 1,027. Basic has the greatest churn volume, but Premium has greater fee exposure because its average monthly fee is higher.

Q4. Are accounts with more profiles associated with different churn rates?

Business goal: Investigate whether the number of profiles on an account is associated with customer retention. If churn rates differ meaningfully, the business could explore whether household usage or account engagement deserves further investigation.


100 % No issues found

Results Messages

	number_of_profiles	total_customers	churned_customers	churn_rate
1	1	972	570	58.640000000000
2	2	1001	575	57.440000000000
3	3	994	575	57.850000000000
4	4	999	375	37.540000000000
5	5	1034	420	40.620000000000

Key finding : Customers with 1–3 profiles show churn rates of approximately 57–59%, compared with around 38–41% for accounts with 4–5 profiles.

This association warrants further investigation into account usage and engagement, but does not prove that adding profiles reduces churn.

Q5.Would our business priorities change if questionable data were excluded?

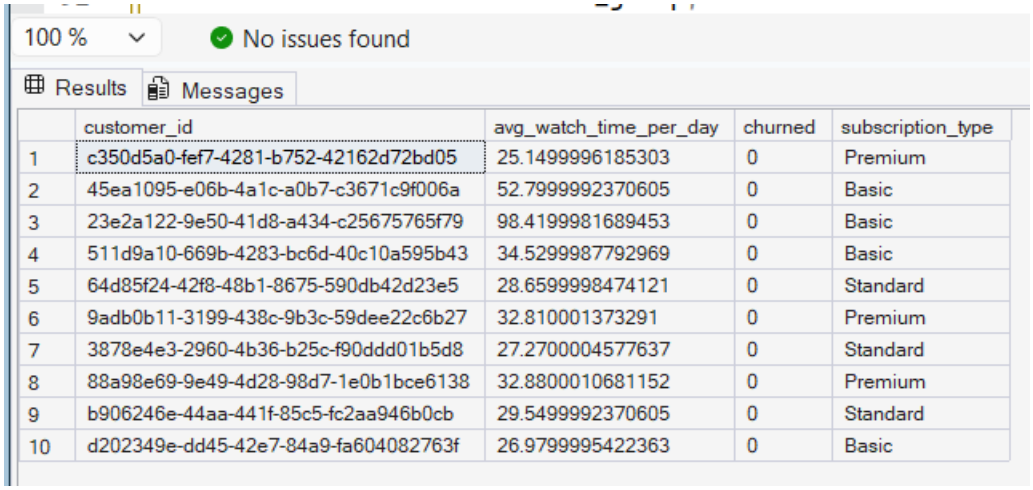
Business goal

Determine whether questionable watch-time records materially affect our churn findings, so that retention recommendations are based on reliable data.


100 % No issues found

Results Messages

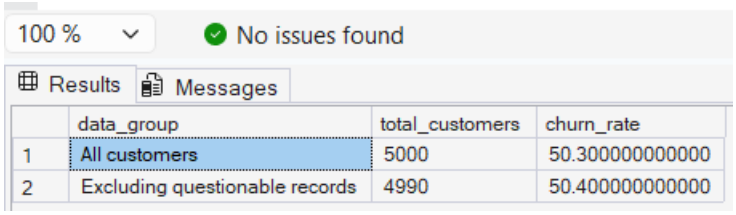
	total_customers	records_above_24_hours
1	5000	10



100 % No issues found

Results Messages

	customer_id	avg_watch_time_per_day	churned	subscription_type
1	c350d5a0-fef7-4281-b752-42162d72bd05	25.1499996185303	0	Premium
2	45ea1095-e06b-4a1c-a0b7-c3671c9f006a	52.7999992370605	0	Basic
3	23e2a122-9e50-41d8-a434-c25675765f79	98.4199981689453	0	Basic
4	511d9a10-669b-4283-bc6d-40c10a595b43	34.5299987792969	0	Basic
5	64d85f24-42f8-48b1-8675-590db42d23e5	28.6599998474121	0	Standard
6	9adb0b11-3199-438c-9b3c-59dee22c6b27	32.810001373291	0	Premium
7	3878e4e3-2960-4b36-b25c-f90ddd01b5d8	27.2700004577637	0	Standard
8	88a98e69-9e49-4d28-98d7-1e0b1bce6138	32.8800010681152	0	Premium
9	b906246e-44aa-441f-85c5-fc2aa946b0cb	29.5499992370605	0	Standard
10	d202349e-dd45-42e7-84a9-fa604082763f	26.9799995422363	0	Basic



100 % No issues found

Results Messages

	data_group	total_customers	churn_rate
1	All customers	5000	50.300000000000
2	Excluding questionable records	4990	50.400000000000

Key finding

- Excluding the 10 records with **daily watch time above 24 hours** reduces the dataset from 5,000 to 4,990 customers.
- The overall churn rate changes only from 50.3% to 50.4%, an increase of 0.1 percentage points.
- This suggests that these questionable records have very little impact on the overall churn rate.
- All 10 records were labelled as not churned, but the exclusion changes the denominator as well as the churn proportion.

Q6 — Is churn associated with payment method?

Business goal: Identify whether churn differs by payment method so the business can investigate potential payment friction or differences in customer behaviour.

100 % No issues found

Results Messages

	payment_method	regions	total_customers	churned_customers	churn_rate
1	Credit Card	Europe, Asia, Oceania, Africa, South America, North America	973	424	43.580000000000
2	Crypto	Oceania, Europe, South America, Africa, North America, Asia	995	594	59.700000000000
3	Debit Card	North America, Oceania, Asia, South America, Europe, Africa	1030	450	43.690000000000
4	Gift Card	Europe, Oceania, Asia, North America, Africa, South America	976	564	57.790000000000
5	PayPal	South America, Oceania, Europe, Africa, Asia, North America	1026	483	47.080000000000

Key finding:

- Crypto: Highest churn rate at 59.70% (594 of 995 customers).
- Gift Card: Second-highest at 57.79% (564 of 976 customers).
- PayPal: Churn rate of 47.08% (483 of 1,026 customers).
- Debit Card: Churn rate of 43.69% (450 of 1,030 customers).
- Credit Card: Lowest churn rate at 43.58% (424 of 973 customers).

Q7. Does churn differ by customers' favorite genre?

Business goal: Determine whether churn rates vary across customers' favorite genres and identify content preferences that may warrant further investigation.

100 % No issues found

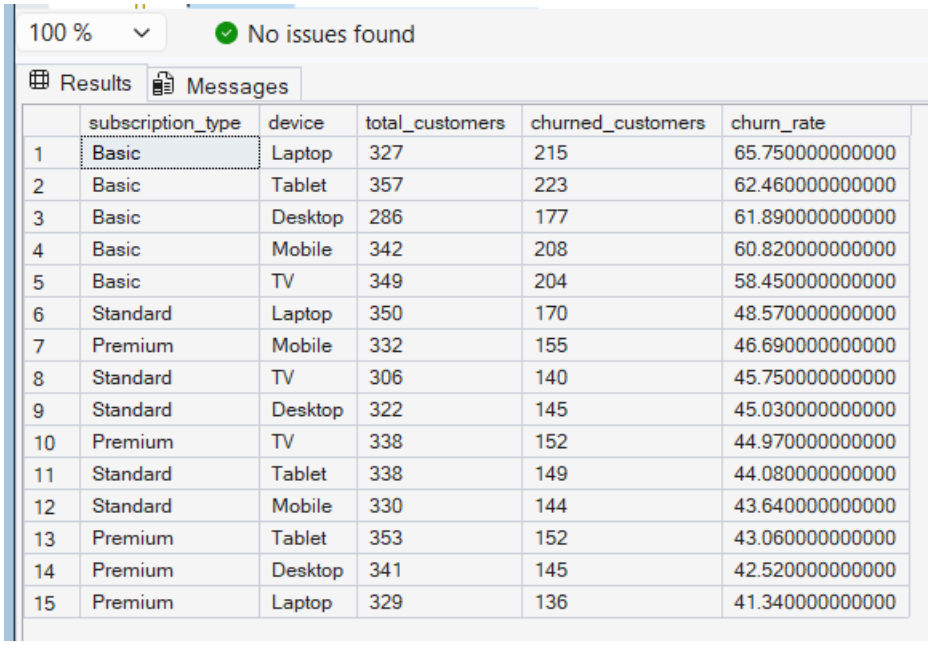
Results Messages

	favorite_genre	total_customers	churned_customers	churn_rate
1	Action	697	365	52.370000000000
2	Drama	731	382	52.260000000000
3	Horror	713	367	51.470000000000
4	Documentary	729	370	50.750000000000
5	Comedy	685	342	49.930000000000
6	Romance	725	350	48.280000000000
7	Sci-Fi	720	339	47.080000000000

Key finding: Action has the highest churn rate (52.37%), closely followed by Drama (52.26%). Sci-Fi has the lowest (47.08%), a difference of 5.29 percentage points.

Q8. Are some device-and-plan combinations associated with unusually high churn?

Business goal: Identify specific subscription-plan and device combinations with higher churn rates so the business can investigate which customer segments may need closer attention.



100 % No issues found

Results Messages

	subscription_type	device	total_customers	churned_customers	churn_rate
1	Basic	Laptop	327	215	65.750000000000
2	Basic	Tablet	357	223	62.460000000000
3	Basic	Desktop	286	177	61.890000000000
4	Basic	Mobile	342	208	60.820000000000
5	Basic	TV	349	204	58.450000000000
6	Standard	Laptop	350	170	48.570000000000
7	Premium	Mobile	332	155	46.690000000000
8	Standard	TV	306	140	45.750000000000
9	Standard	Desktop	322	145	45.030000000000
10	Premium	TV	338	152	44.970000000000
11	Standard	Tablet	338	149	44.080000000000
12	Standard	Mobile	330	144	43.640000000000
13	Premium	Tablet	353	152	43.060000000000
14	Premium	Desktop	341	145	42.520000000000
15	Premium	Laptop	329	136	41.340000000000

Key findings:

- **Basic + Laptop**: Highest churn at 65.75% (215 of 327 customers).
- **Basic + Tablet:** 62.46% churn (223 of 357 customers).
- **Basic + Desktop:** 61.89% churn (177 of 286 customers).
- **Premium + Laptop:** Lowest churn at 41.34% (136 of 329 customers).
- All five Basic-plan device groups have churn rates above 58%, while Premium-plan groups range from 41.34% to 46.69%.